
CSE471

DEEP LEARNING FOR COMPUTER VISION

Lecture #0

A kick start session

Course details

- **LTP – 2 - 0 - 2**
- **Credits – 3.0**
- **Text Books:**
 - FUNDAMENTALS OF DEEP LEARNING & COMPUTER VISION by NIKHIL SINGH, PARAS AHUJA, BPB PUBLICATIONS

References:

- DEEP LEARNING by AMIT KUMAR DAS, SAPTRISHI GOSWAMI, PABITRA MITTRA, AMLAN CHAKRABARTI, Pearson Education India
- ELEMENTS OF DEEP LEARNING FOR COMPUTER VISION by BHARAT SIKKA, BPB PUBLICATIONS
- DEEP LEARNING FOR COMPUTER VISION: EXPERT TECHNIQUES TO TRAIN ADVANCED NEURAL NETWORKS USING TENSORFLOW AND KERAS by RAJALINGAPPAA SHANMUGAMANI, PACKT PUBLISHING



School Vision

- To be a globally recognized school through excellence in teaching, learning and research for creating Computer Science professionals, leaders and entrepreneurs of future contributing to society and industry for sustainable growth.



School Mission

- To build computational skills through hands-on and practice-based learning with measurable outcomes.
- To establish a strong connect with industry for in-demand technology driven curriculum.
- To build the infrastructure for meaningful research around societal problems.
- To nurture future leaders through research-infused education and lifelong learning.
- To create smart and ethical professionals and entrepreneurs who are recognized globally.

Course Outcomes

- CO1 :: remember the fundamental concepts of computer vision, image representation, preprocessing operations, and neural network basics used in deep learning.
- CO2 :: understand the working principles of convolutional neural networks, feature extraction, pooling, regularization, and CNN training techniques.
- CO3 :: apply transfer learning and fine-tuning methods using pre-trained models for domain specific computer vision tasks.
- CO4 :: apply object detection and localization techniques, including bounding box representation, IoU, NMS, and modern detection architectures.
- CO5 :: apply image segmentation methods and transformer-based vision architectures for extracting meaningful visual information from images.
- CO6 :: understand the principles of generative vision models, vision-language models, and interpretability techniques used in advanced computer vision applications

Program Outcomes

PO1::Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2::Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PO3::Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4::Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5::Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

Program Outcomes (Cont.)

PO6::Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7::Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8::Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9::Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10::Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.



Program Outcomes (Cont.)

PO11: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Recognise the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.



Program Specific Outcomes

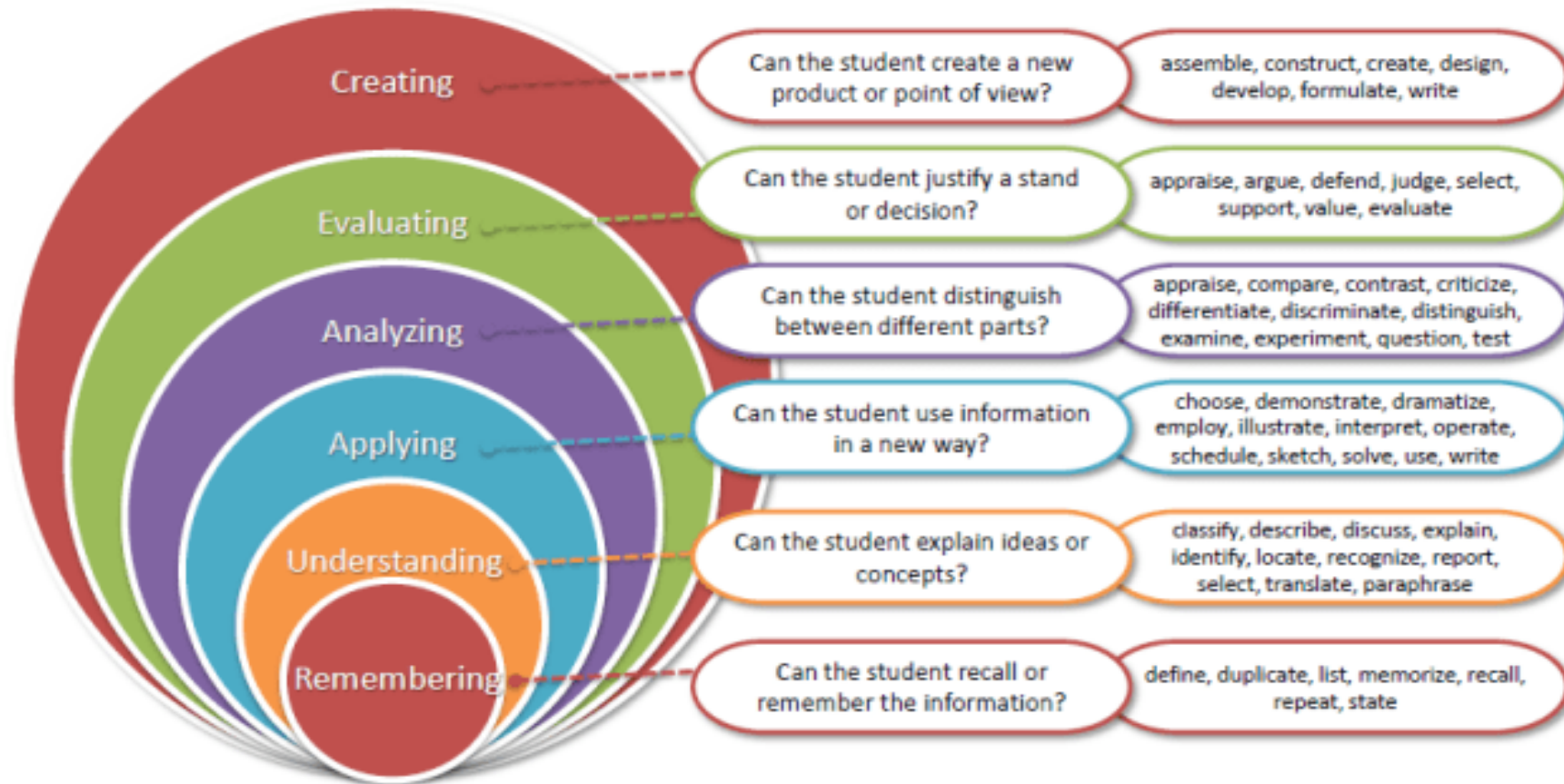
PSO1::Apply basic knowledge in the areas such as Software Engineering, Networking and Security, Database Management Systems, Intelligent Systems, Operating Systems and System Architecture for building Software products.

PSO2::Provide effective and efficient real time solutions using attained knowledge in inter-disciplinary domains for societal benefits through projects.

CO-PO Mapping

CO / PO & PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	–	1	2	–	–	–	–	2	–	1	2	1
CO2	3	3	1	2	3	–	–	–	–	1	–	2	3	2
CO3	3	3	1	2	3	–	–	–	–	1	–	2	3	2
CO4	2	3	2	2	3	–	–	1	–	2	–	2	3	2
CO5	3	3	2	3	3	–	–	–	1	2	1	2	3	3
CO6	3	3	3	2	3	1	–	1	2	3	2	3	3	3

Revised Bloom's Taxonomy





Before
MTE

Course contents

UNIT-I

- **Foundations of Computer Vision and Deep Learning** : Role of computer vision in modern AI applications, Traditional versus deep-learning-based vision approaches, image fundamentals, pixels, channels, color spaces, resizing, cropping and normalization, neural network basics, perceptron, feedforward networks, activation functions, loss functions, optimization methods



Before
MTE

UNIT-II

- **Convolutional Neural Networks and Training Techniques** : CNN Basics: convolutional neural networks, convolution operation, filters, feature maps, padding, stride, pooling, and receptive fields, Modern CNN architectures: LeNet, AlexNet, VGG, ResNet, and Inception networks, regularization techniques, batch normalization and dropout, data augmentation methods, learning rate scheduling, weight initialization and early stopping



Before
MTE

UNIT-III

- **Transfer Learning and Fine-Tuning for Vision Tasks** : Pre-trained architecture: transfer learning, ResNet, EfficientNet, MobileNet, feature extraction, fine-tuning, managing freezing and unfreezing of layers, applying layer-wise learning rates, domain adaptation and dataset



UNIT-IV

- **Object Detection and Localization** : object detection: object detection principles ,algorithms, bounding box representation, intersection over union, non-maximum suppression, classical detection pipelines, RCNN, single-stage detectors, YOLO variants, SSD, RetinaNet with focal loss, Evaluation metrics: mean average precision, IoU thresholds



After
MTE

UNIT-V

- **Image Segmentation and Advanced Vision Architectures** : Image Segmentation: semantic segmentation, instance segmentation, UNet, DeepLab architectures, instance segmentation using Mask R-CNN, Vision transformers: self-attention, multi-head attention, positional encoding, transformer-based models, ViT, Swin Transformer



UNIT-VI

- **Generative Vision Models** : Generative deep learning: variational autoencoders, GAN architectures, DCGAN, CycleGAN, StyleGAN, image generation, image-to-image translation, super-resolution, vision language models: CLIP for image-text alignment, interpretability techniques, Grad-CAM, saliency maps.

List of Practical's:

1. Load an image using OpenCV, perform resizing, grayscale conversion, normalization, and visualize the results using Matplotlib in multiple subplots.
2. Implement a 2D convolution operation using NumPy, apply it to a grayscale image patch with different filters, and visualize the resulting feature maps.
3. Implement zero padding and stride 2D convolution using NumPy on a grayscale image patch. Compare the output feature maps obtained with different padding and stride values.
4. Train LeNet and ResNet-18 on MNIST and compare training time, accuracy, and model size to understand how architectural depth and skip connections influence performance.
5. Use a pretrained to extract image features and train a classifier on a new dataset such as CIFAR-10 or a small custom dataset (e.g., flower images). Evaluate the classifier's accuracy.



List of Practical's (Cont.) :

6. Fine-tune a pretrained MobileNet model on a small dataset of Cats vs Dogs. Experiment with freezing/unfreezing layers and observe the effect on training time and accuracy.
7. Use a pretrained YOLO model to detect objects in images or webcam/video streams, draw bounding boxes and class labels, and analyze detection confidence and performance
8. Fine-tune a pretrained Faster R-CNN model on a small custom dataset . Train for a few epochs and evaluate performance.
9. Use a pretrained UNet model to generate pixel-wise segmentation masks for sample images, overlay the predicted mask on the original image, and analyze the model's ability to segment objects
10. Use a pretrained CNN model to classify images, then apply Grad-CAM to visualize which parts of the image contributed most to the prediction. Compare Grad-CAM outputs for correctly classified and misclassified samples.



Course Assessment Model

- **Marks break up**

- Attendance

5

- Continuous Assessment

45

- ETT

+ 50

- **Total**

100



Details of Academic tasks

Academic Task	Objective	Detail of Academic Task	Nature of Task	Mode of Academic Task	Max Marks	Allotment Week
Test 1	Student will be able to learn the basic concepts of computer vision and deep learning along with Convolution Neural Network architecture	Unit-I and Unit-II will be covered	Individual	Offline	30	5/6
Project (compulsory)	The objective of this project is to apply complete deep learning for computer vision to solve real world problems by designing, implementing and evaluating the techniques.	Students are required to carry out a comprehensive computer vision project by covering the entire course syllabus, involving problem identification, dataset selection, text preprocessing, model design and implementation using deep learning and transformer-based approaches, performance evaluation using suitable metrics, and analysis of results, followed by proper documentation and presentation of the work.	Group	Online	30	3/11
Test 2	To test the subject knowledge of the students and provide opportunity to improve their CA performance in case of low marks or any missed CA	Subjective test covering syllabus from Unit-1 to Unit-IV	Individual	Offline	30	11/12

Note: The project is compulsory (weightage 67%) and One best out of Test-1 and Test-2 will be considered with weightage 33% in final CA calculation.

OPEN EDUCATIONAL RESOURCE

Course Code	Course Title	Unit mapped	Broad topic	OER Type	Title of OER	*%age unit mapped with OER (approx)	Source URL
INT472 : Deep Learning for Natural Language Processing		Unit 1	Foundations of NLP, Language & Grammar, Morphology, Syntax, Semantics, Tokenization, Stemming, Lemmatization, Stop-word Removal, Bag-of-Words, N-grams, TF-IDF	Course	DeepLearning.AI Natural Language Processing Specialization (Course 1), NLTK Book, Google Machine Learning Text Classification Guide	95%	https://www.coursera.org/specializations/natural-language-processing https://www.nltk.org/book/ https://developers.google.com/machine-learning/guides/text-classification
		Unit 2	Vector Space Models, Word Embeddings, Word2Vec, CBOW, Skip-Gram, GloVe, Semantic Similarity, PCA, t-SNE	Course	DeepLearning.AI NLP Specialization (Course 2), Stanford CS224N Notes, Illustrated Word2Vec, GloVe Project	95%	https://www.coursera.org/specializations/natural-language-processing https://web.stanford.edu/class/cs224n/ https://jalamar.github.io/illustrated-word2vec/ https://nlp.stanford.edu/projects/glove/

OPEN EDUCATIONAL RESOURCE

Course Code	Course Title	Unit mapped	Broad topic	OER Type	Title of OER	*%age unit mapped with OER (approx)	Source URL
INT344 : Natural Language Processing		Unit 3	Deep Learning Sequence Models for NLP : sequential text data, recurrent neural networks, long short term memory networks, gated recurrent units, bidirectional RNNs, sequence modeling applications, sentiment classification, text classification, sequence training techniques, teacher forcing, truncated backpropagation through time, evaluation metrics for sequence tasks	Course	Natural Language Processing Specialization (Course 2): Dive into Deep Learning (D2L), Stanford CS224N: NLP with Deep Learning, Speech and Language Processing (3rd Edition Draft)	95%	https://www.coursera.org/specializations/natural-language-processing https://d2l.ai/chapter_recurrent-neural-networks/index.html https://web.stanford.edu/class/cs224n/

Course Code	Course Title	Unit mapped	Broad topic	OER Type	Title of OER	*%age unit mapped with OER (approx)	Source URL
INT344 : Natural Language Processing		Unit 4	Sequence-to-Sequence Models and Attention Mechanisms : encoder–decoder architectures for	Course	Natural Language Processing Specialization (Course 1), Jay Alammarr's Illustrated Guides, Stanford CS224N: NLP with Deep Learning	90%	https://www.coursera.org/specializations/natural-language-processing
			NLP, sequence-to-sequence models for machine translation and summarization, attention in deep				https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/
			NLP, soft attention, alignment mechanisms, Bahdanau and Luong attention, integrating attention into				https://web.stanford.edu/class/cs224n/
			encoder–decoder networks, evaluation techniques, BLEU and ROUGE scores, limitations of classical				https://classic.d2l.ai/chapter_attention-mechanisms/index.html
			seq2seq models				

OPEN EDUCATIONAL RESOURCE

Course Code	Course Title	Unit mapped	Broad topic	OER Type	Title of OER	*%age unit mapped with OER (approx)	Source URL
INT344 : Natural Language Processing		Unit 5	Transformers : transformer architecture, self-attention, multi-head attention, positional encoding,	Course	Natural Language Processing Specialization (Course 3): Sequence Models, CS224N: Deep Learning for NLP	95%	https://huggingface.co/learn/nlp-course/
			transformer encoder and decoder blocks, tokenization methods, Byte-Pair Encoding and WordPiece				https://jalammar.github.io/illustrated-transformer/
			Pretrained Language Models : pretrained transformer models, BERT, GPT, T5, masked language				https://web.stanford.edu/class/cs25/
			modeling, next sentence prediction, causal language modeling, transfer learning for NLP tasks, fine				https://huggingface.co/docs/transformers/index
			tuning for text classification, named entity recognition,				

OPEN EDUCATIONAL RESOURCE

Course Code	Course Title	Unit mapped	Broad topic	OER Type	Title of OER	*%age unit mapped with OER (approx)	Source URL
INT344 : Natural Language Processing		Unit 6	Generative NLP and LLMs : generative NLP models, text generation strategies, greedy search, beam search, top-k, nucleus sampling, instruction-tuned, large language models, model behaviors in	Course	Natural Language Processing Specialization (Course 4): Attention Models, Hugging Face Course, Transformer Models	95%	https://huggingface.co/learn/nlp-course/
			summarization, dialogue generation, reasoning tasks, evaluation metrics, perplexity and human judgment measures, explainability and hallucination in LLMs				https://www.cloudskillsboost.google/journeys/118 https://www.coursera.org/learn/generative-ai-with-llms https://cohere.com/llmu https://www.youtube.com/watch?v=35X6zlhoCy4

EDU Revolution



- EDU Revolution is LPU's student-centric academic framework that recognizes and rewards learning beyond the classroom.
- Students gain academic recognition, industry exposure, and career-ready skills through:
 - Revenue Generation opportunities,
 - Projects & Hackathons,
 - National and International Internship Opportunities, and
 - Certifications

EDU Revolution Initiatives for CSE471

Category	Description
Co-curricular/ Extra curricular activities	Students may be considered for CA exemption if they: • Secure a position/rank in Smart India Hackathon or any recognized hackathon/innovation competition, and Work on a problem statement substantially related to COMPUTER VISION or AI-based vision technologies requiring concepts covered in this course. • The project/problem statement should involve areas such as: object classification, clustering, segmentation, detection from images/videos. • Students are expected to secure a position among the top three- • Winner, • 1st Runner-Up, or • 2nd Runner-Up

Students can apply for various EDU Revolution Initiatives through the following link:

- **Login UMS --> LMS---> Edu Revolution: Be the Change---> Apply for EDU Revolution Edu-Revolution**

Query and Assistance Zone: for guidance and support on EDU Revolution students can visit Block 38-205B

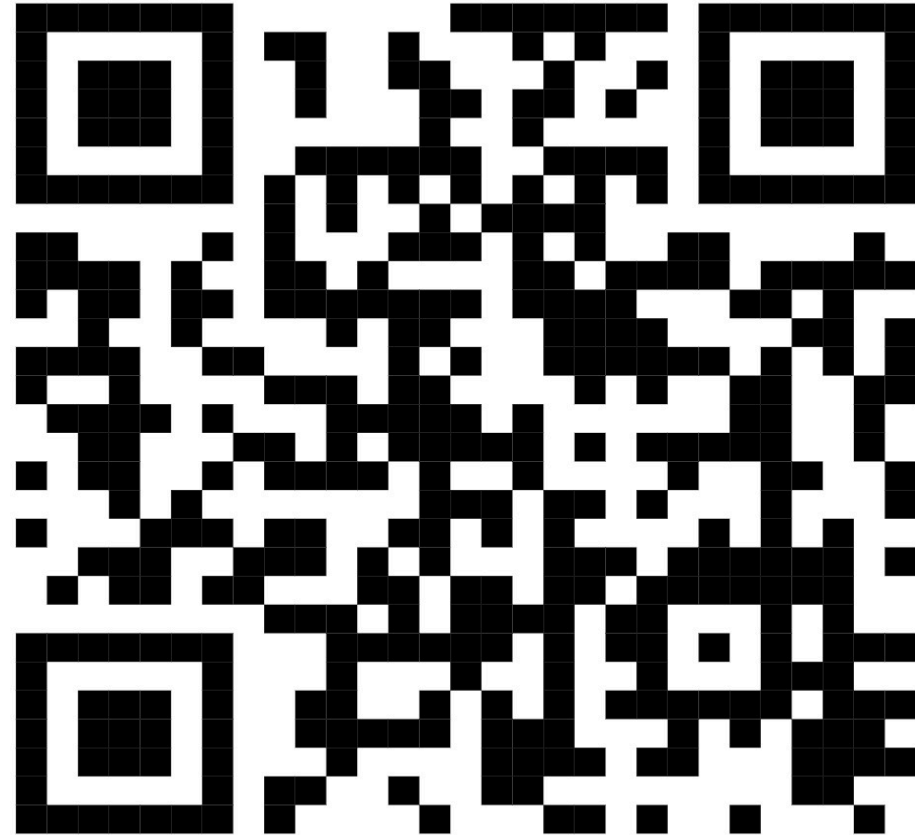


Lecture-0 Feedback

- Scan this QR-Code to leave your feedback about Lecture-0

Or
Visit URL:

<https://forms.gle/pksSDrADkNLde4dBA>





Any Queries.....